Shun/修海 (@shun4midx)

SUMS OF INDEPENDENT DISCRETE RANDOM VARIABLES

PROPOSITION

Let X,Y be indep discrete random variables w/ prob mass funcs P_X, P_Y.
Then, their sum has PMF $P_{X+Y}(z)=\sum_x P_X(x)P_Y(z-x)=\sum_y P_X(z-y)P_Y(y)$

Proof

Notice, $P_{X+Y}(z)=\mathbb{P}(X+Y=z)$

We consider $E=\{X+Y=z\}$
The sample space S can be partitioned based on the value of x
Define $F_i:=\{X=x_i\}$, where $x_1,x_2,\dots$ are the vals X takes

By the Law of Total Prob,

$$\begin{aligned}\mathbb{P}(X+Y=z)&=\sum_i \mathbb{P}(X+Y=z\mid X=x_i)\mathbb{P}(X=x_i)\\&=\sum_i \mathbb{P}(Y=z-x_i\mid X=x_i)\mathbb{P}(X=x_i)\\ \text{(independence)}&=\sum_i \mathbb{P}(Y=z-x_i)\mathbb{P}(X=x_i)\\&=\sum_i P_X(x_i)P_Y(z-x_i)\\&=\sum_x P_X(x)P_Y(z-x)\ \square\end{aligned}$$

EXAMPLE

Sums of Poisson random variables
Suppose $X\sim Poi(\lambda_1)$, $Y\sim Poi(\lambda_2)$ are indep
Then $X+Y\sim ?$

Solution

nonneg int

$$P_{X+Y}(z)=\sum_{x=0}^{z}P_X(x)P_Y(z-x)=\sum_{x=0}^{z}\frac{\lambda_1^x e^{-\lambda_1}}{x!}\frac{\lambda_2^{z-x}e^{-\lambda_2}}{(z-x)!}=e^{-(\lambda_1+\lambda_2)}\sum_{x=0}^{z}\frac{\lambda_1^x\lambda_2^{z-x}}{x!(z-x)!}=\frac{e^{-(\lambda_1+\lambda_2)}}{z!}\sum_{x=0}^{z}\binom{z}{x}\lambda_1^x\lambda_2^{z-x}\overset{\text{bi. thm}}{=}\frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^z}{z!}$$

$\therefore$ $X+Y\sim Poi(\lambda_1+\lambda_2)$

CONDITIONAL RANDOM VARIABLES

SETTING

Two rand vars in a prob space, not necessarily indep.
If we know the val of one, what does that tell us about the other?

DISCRETE CASE

DEFINITION

Let X,Y be discrete rand vars.
The conditional prob mass function of X given $Y=y$ is given by $P_{X|Y}(x|y)=\mathbb{P}(X=x\mid Y=y)$

By def of cond probs, $P_{X|Y}(x|y)=\mathbb{P}(X=x\mid Y=y)=\frac{\mathbb{P}(X=x,Y=y)}{\mathbb{P}(Y=y)}=\frac{P_{X,Y}(x,y)}{P_Y(y)}$

If X,Y are indep, then $P_{X,Y}(x,y)=P_X(x)P_Y(y)$, so $P_{X|Y}(x|y)=P_X(x)$

EXAMPLE

$X\sim Poi(\lambda_1)$, $Y\sim Poi(\lambda_2)$ indep, what is the conditional dist of X given $X+Y=n$?

Solution

$$P_{X|X+Y}(x|n)=\mathbb{P}(X=x\mid X+Y=n)=\frac{\mathbb{P}(X=x,X+Y=n)}{\mathbb{P}(X+Y=n)}=\frac{\mathbb{P}(X=x,Y=n-x)}{\mathbb{P}(X+Y=n)}\overset{\text{indep}}{=}\frac{\mathbb{P}(X=x)\mathbb{P}(Y=n-x)}{\mathbb{P}(X+Y=n)}$$

$$\overset{\text{from earlier}}{=}\frac{\left(\frac{\lambda_1^x e^{-\lambda_1}}{x!}\right)\left(\frac{\lambda_2^{n-x}e^{-\lambda_2}}{(n-x)!}\right)}{\frac{(\lambda_1+\lambda_2)^n e^{-(\lambda_1+\lambda_2)}}{z!}}=\binom{n}{x}\left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^x\left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^{n-x}$$

This is the PMF of a bin rand var
$\therefore$ X cond on $X+Y=n\sim Bin\left(n,\frac{\lambda_1}{\lambda_1+\lambda_2}\right)$

EXAMPLE

We repeat a random experiment n times, indep, with prob of success $=p$.
Let $X = \#$ success, so $X \sim Bin(n,p)$
Let $\sigma =$ seq of outcomes (success/failure) $\in \{S,F\}^n$

What is the distribution of σ given $X=k$?

Solution

$$P_{\sigma|X}(s|k) = \frac{P_{\sigma,X}(s,k)}{P_X(k)} = \frac{\mathbb{P}(\sigma=s, X=k)}{\mathbb{P}(X=k)}$$

$$= \frac{p^k(1-p)^k}{\binom{n}{k}p^k(1-p)^k}$$

$$= \frac{1}{\binom{n}{k}}$$

CONTINUOUS RANDOM VARIABLES

X, Y are cont. rand vars in a prob space. If $Y=y$, what can we say about X?
It doesn't make sense to consider $\mathbb{P}(X=x|Y=y)$ directly.

$$\mathbb{P}(x \le X \le x+\delta_x | y \le Y \le y+\delta_y)$$

$$= \frac{\mathbb{P}(x \le X \le x+\delta_x, y \le Y \le y+\delta_y)}{\mathbb{P}(y \le Y \le y+\delta_y)} = \frac{\int_x^{x+\delta_x}\int_y^{y+\delta_y} f_{X,Y}(u,v)\,dv\,du}{\int_y^{y+\delta_y} f_Y(v)\,dv} \approx \frac{f_{X,Y}(x,y)\,\delta_x\,\delta_y}{f_Y(y)\,\delta_y} \xrightarrow{\delta_x,\delta_y \to 0} \frac{f_{X,Y}(x,y)}{f_Y(y)}dx$$

DEFINITION

The conditional probability density function of X given $Y=y$ is

$$f_{X|Y}(x,y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} \quad \text{and} \quad \mathbb{P}(X \in A | Y=y) = \int_A f_{X|Y}(x,y)\,dx$$

REMARK

If X, Y are indep, then $f_{X|Y}(x|y) = f_X(x) \Rightarrow \mathbb{P}(X \in A | Y=y) = \mathbb{P}(X \in A)$

EXAMPLE

Let $f_{X,Y}(x,y) = \begin{cases} \frac{e^{-\frac{x}{y}}e^{-y}}{y}, & x,y>0 \\ 0, & o/w \end{cases}$

What is $\mathbb{P}(X>1 | Y=y)$?

Solution

$$f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,dx = \begin{cases} 0, & y \le 0 \\ \int_0^\infty \frac{e^{-\frac{x}{y}}e^{-y}}{y}dx = e^{-y}, & y>0 \end{cases}$$

$$\therefore f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{\frac{e^{-\frac{x}{y}}e^{-y}}{y}}{e^{-y}} = \frac{e^{-\frac{x}{y}}}{y} \quad \text{for } x,y>0$$

$$\therefore \mathbb{P}(X>1|Y=y) = \int_1^\infty f_{X|Y}(x|y)\,dx = \int_1^\infty \frac{e^{-x/y}}{y}dx = e^{-\frac{1}{y}}$$

MIXED CASE

Suppose X: cont. rand var and N: discrete rand var in the same probability space
What is the conditional dist of X given $N=n$?

$$\mathbb{P}(x \le X \le x+\delta_x | N=n) = \frac{\mathbb{P}(x \le X \le x+\delta_x, N=n)}{\mathbb{P}(N=n)} = \frac{\mathbb{P}(N=n | x \le X \le x+\delta_x)\,\mathbb{P}(x \le X \le x+\delta_x)}{\mathbb{P}(N=n)} \xrightarrow{\delta_x \to 0} \frac{\mathbb{P}(N=n|X=x)\,\mathbb{P}(X=x)}{\mathbb{P}(N=n)}$$

EXAMPLE

Shun/修海 (@shun4midx)

We take a coin w/ $\mathbb{P}(\text{heads}) = P \sim \text{Unif}([0,1])$

We toss the coin n times and observe k heads.

Given this information, what can we say about the distribution of P?

Solution

$$f_{P|\underbrace{N}_{\#\text{ heads}}}(p|k) = \frac{\mathbb{P}(N=k \mid P=p)\, f_P(p)}{\mathbb{P}(N=k)} \overset{N|P=p \sim \text{Bin}(n,p)}{=} \frac{\binom{n}{k} p^k (1-p)^{n-k} \cdot 1}{\mathbb{P}(N=k)} = Cp^k(1-p)^{n-k} \text{ for some const } C$$

Since $\int_0^1 f_{P|N}(p|k)\,dp = 1$, we can calculate c.